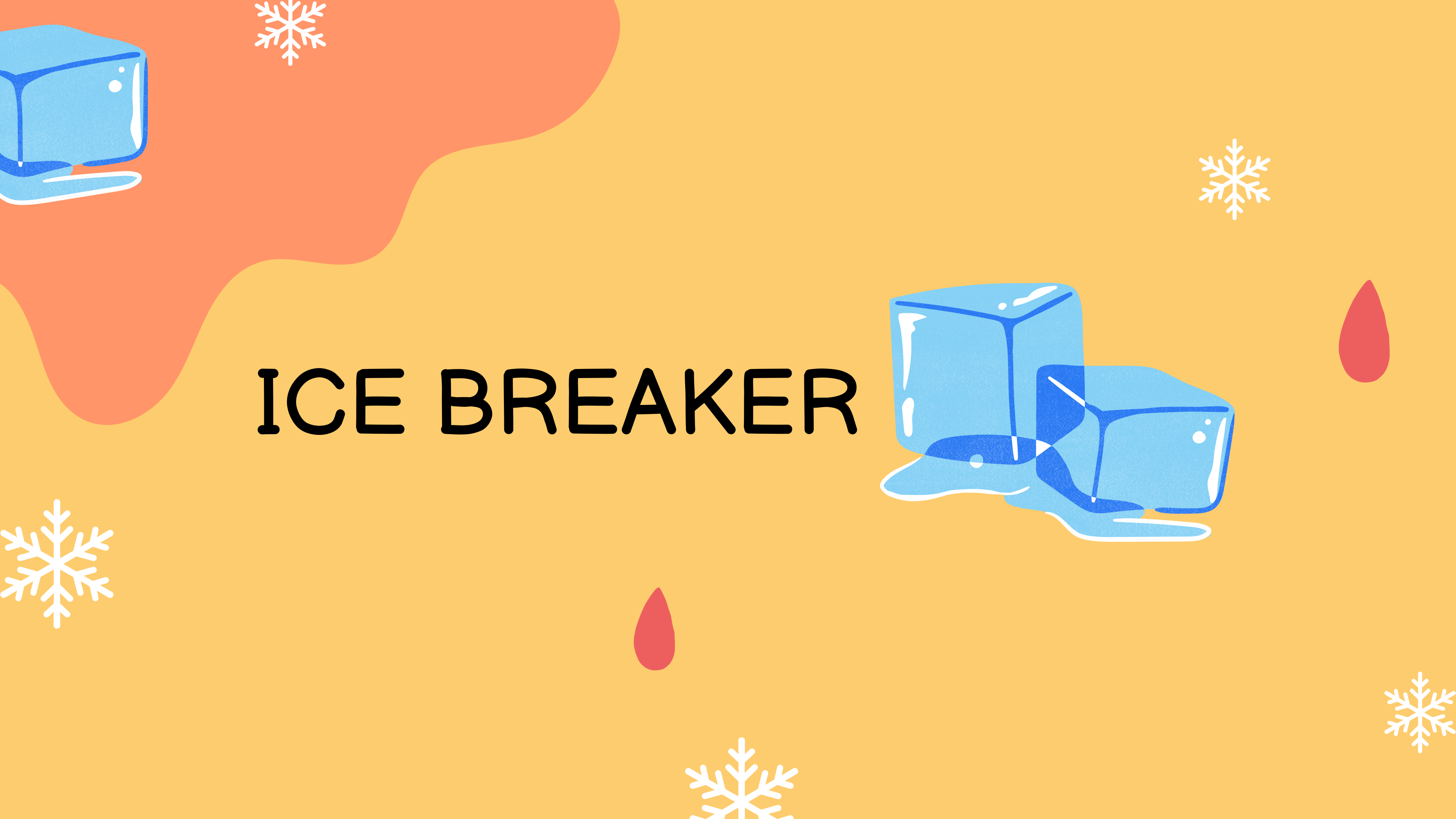
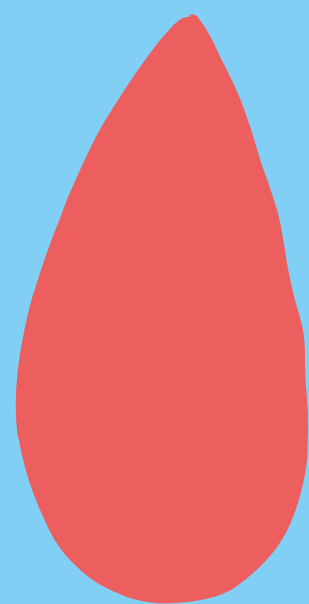


# ICE BREAKER



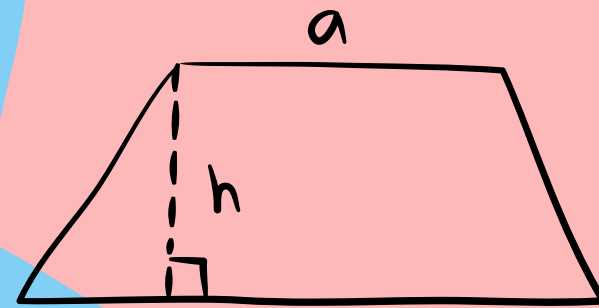
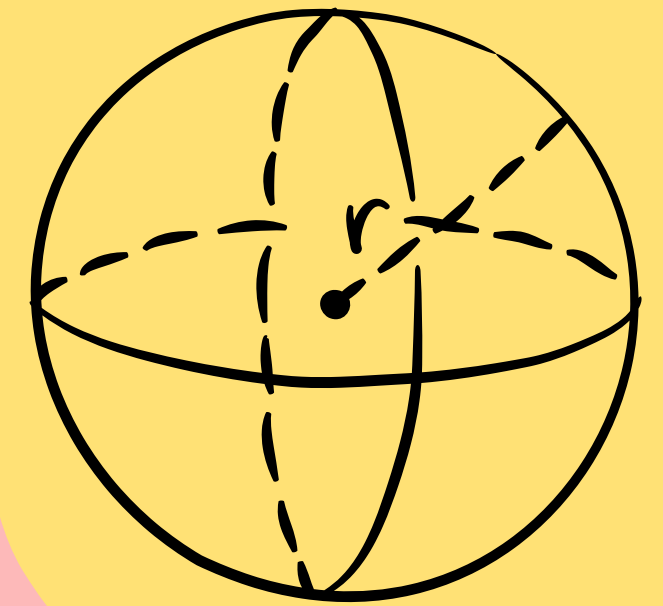
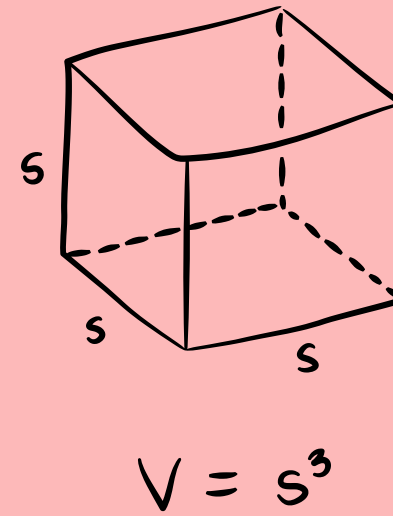
LET US DIVIDE TO FOUR  
GROUPS



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$ax + by = c$$

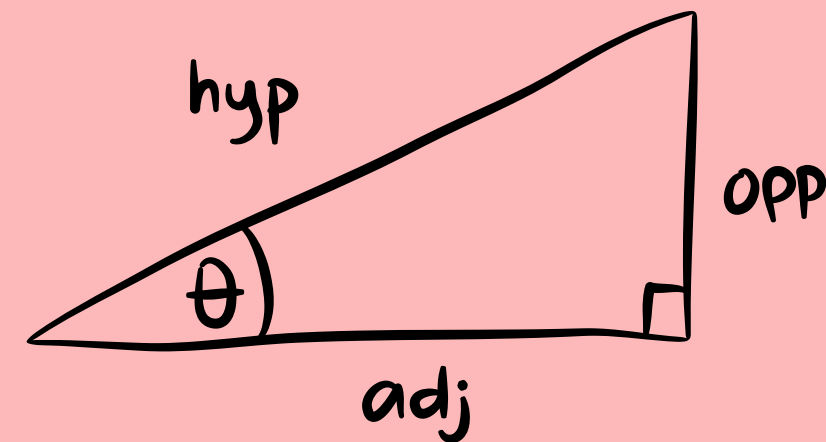
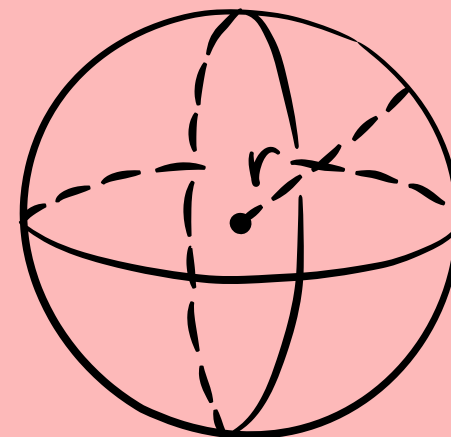
$$ax^2 + bx + c = 0$$



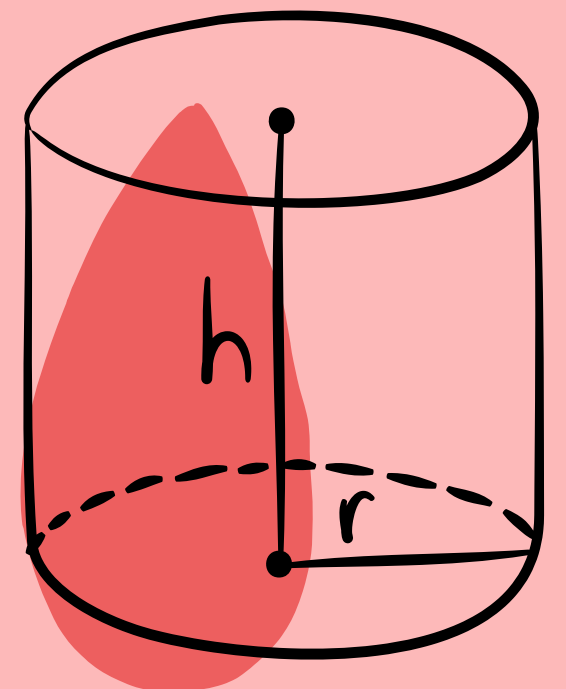
$$A = \frac{a+b}{2} h$$

Write me down:  
Engineering constant  
edition

$$\frac{x}{a} + \frac{y}{b} = 1$$



$$\sin(\theta) = \frac{\text{opp}}{\text{hyp}}$$



$$V = \pi r^2 h$$

$$ax^2 + bx + c = 0$$

# Universal gas constant

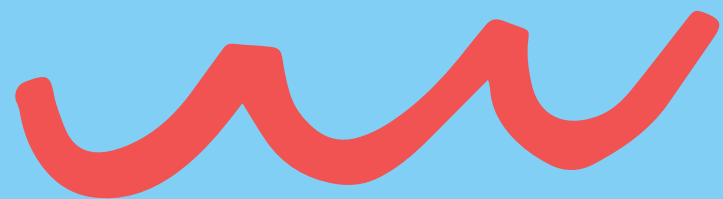
$$8.314 \frac{\text{J}}{\text{mol K}}$$



# Avogadro's number



$$6.022 \times 10^{23} \text{ / mol}$$



# Speed of light

$$3 \times 10^8 \frac{\text{m}}{\text{s}}$$



# Acceleration due to gravity

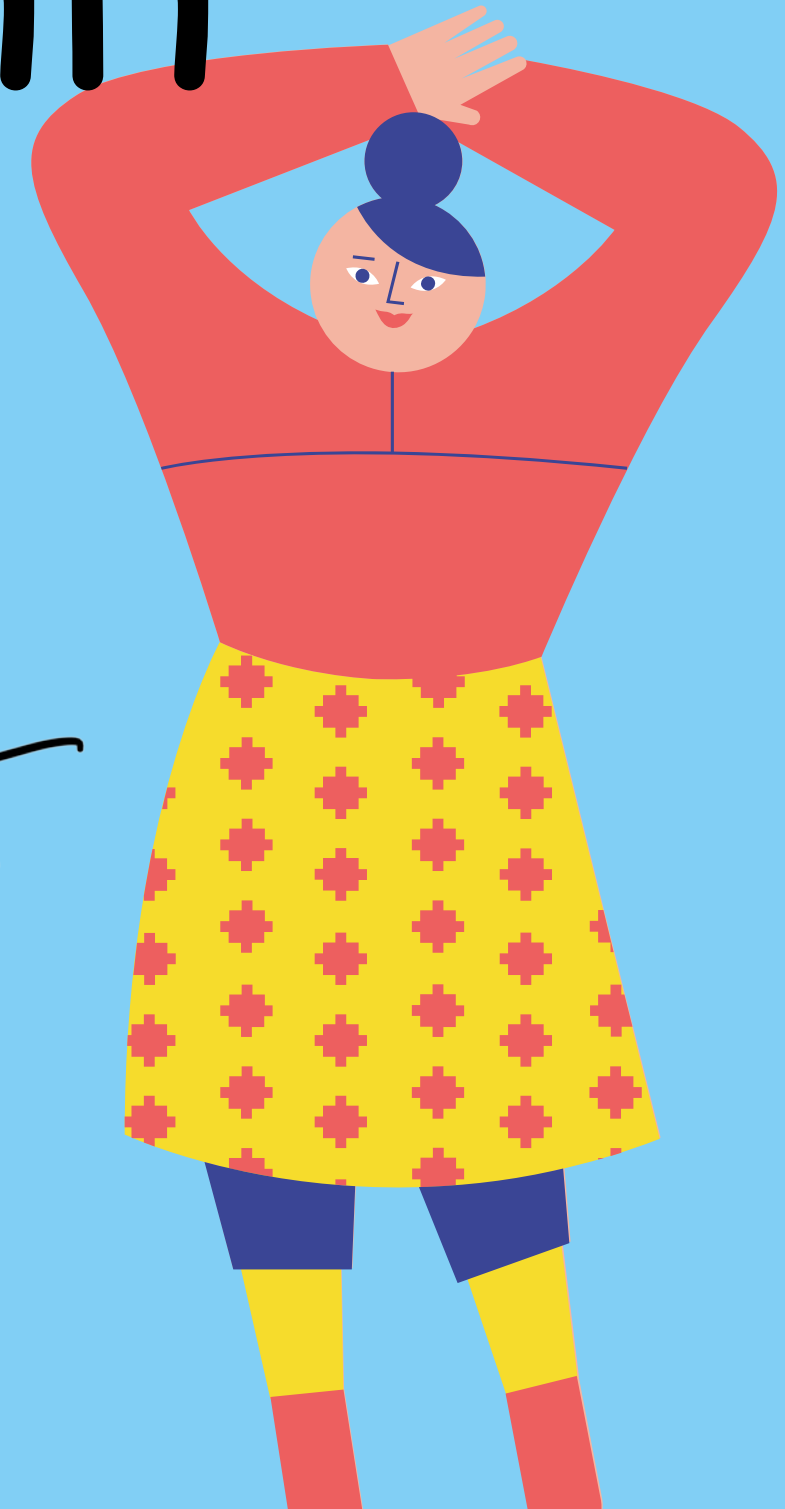
$$9.8 \text{ m/s}^2$$



# Stefan Boltzmann

(metric)

$$5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}$$



# Specific Heat of Water



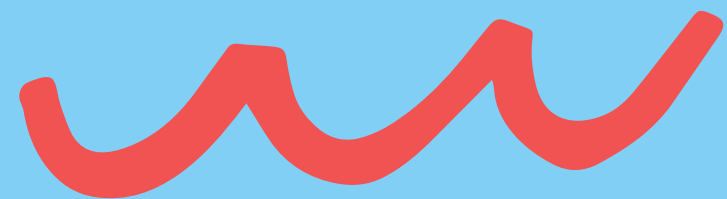
$$4.186 \frac{\text{J}}{\text{g}^\circ\text{C}}$$

# Density of water

1000 kg/m<sup>3</sup>



# Stefan Boltzmann constant (english)



Latent heat of  
vaporization of  
water

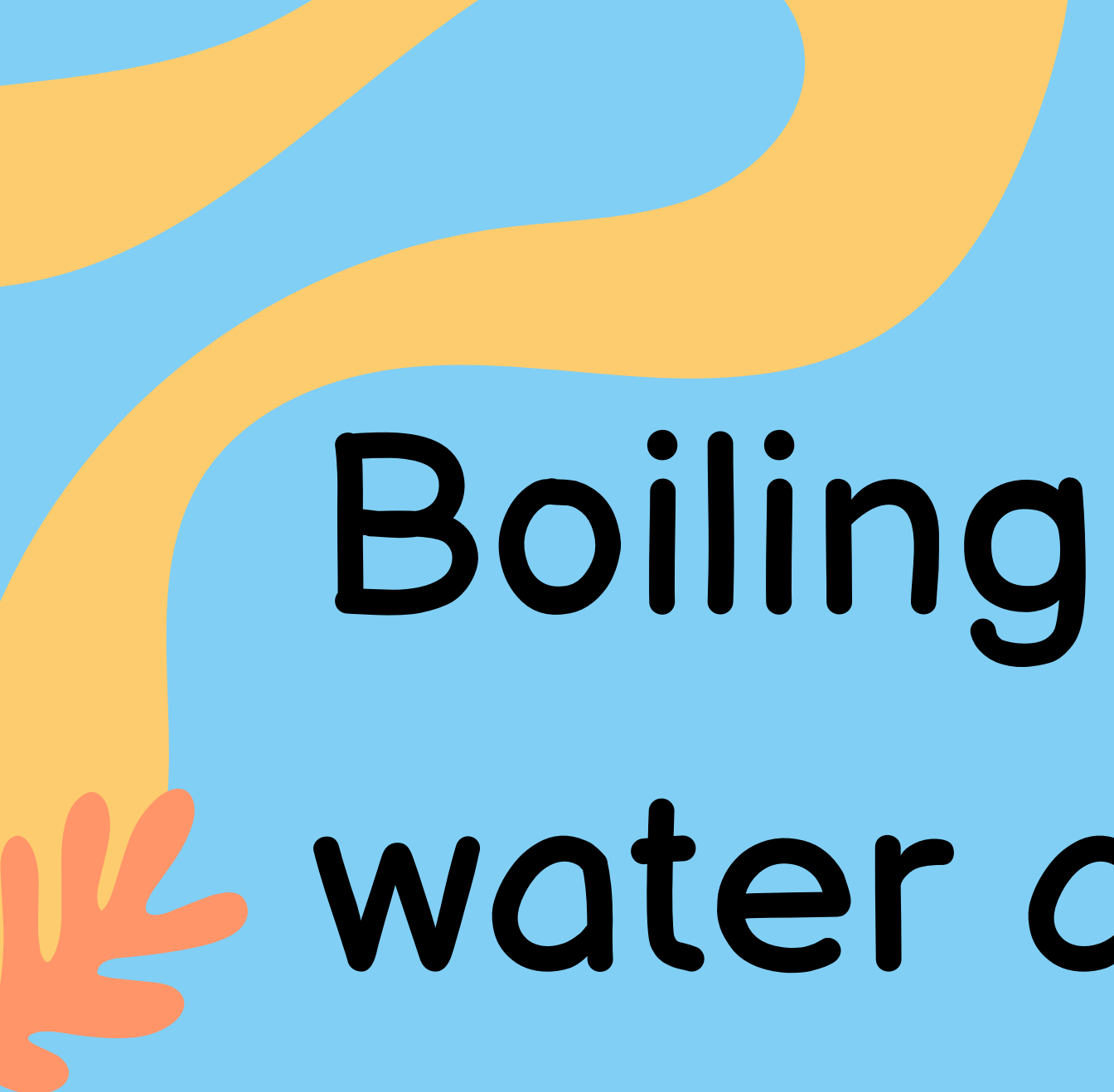
2260 kJ/kg



# Latent heat of fusion (ice)

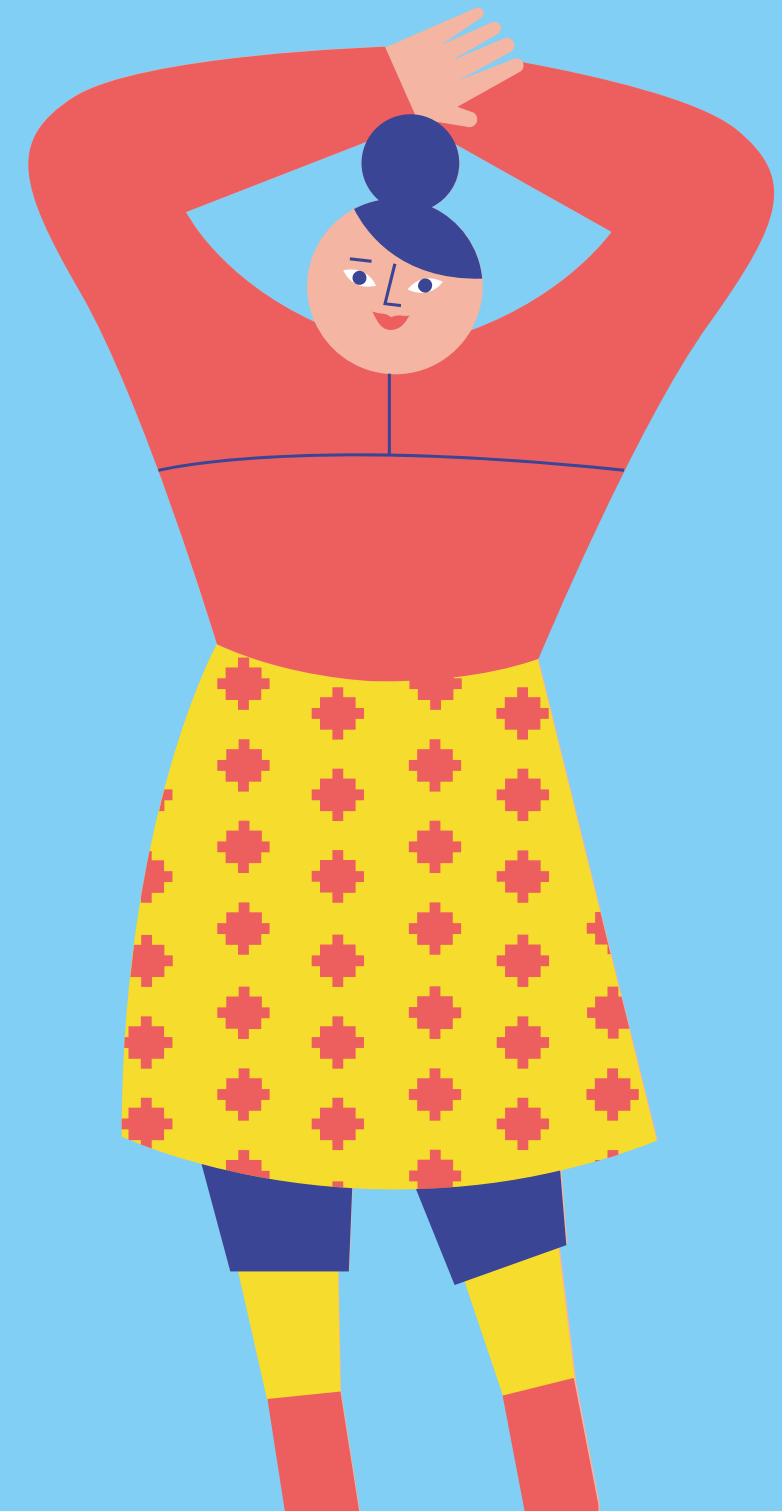
334 kJ/kg





Boiling point of  
water at 1 atm

100°C



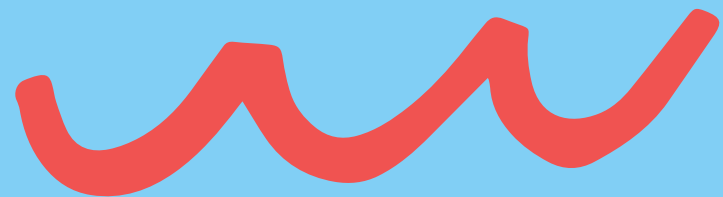
# Freezing point of water



$0^{\circ}\text{C}$  4/92R

# Density of air

$$1.2 \text{ kg/m}^3$$



# Specific heat of dry

air

$$1.002 \frac{\text{KJ}}{\text{Kg} \cdot ^\circ\text{C}}$$



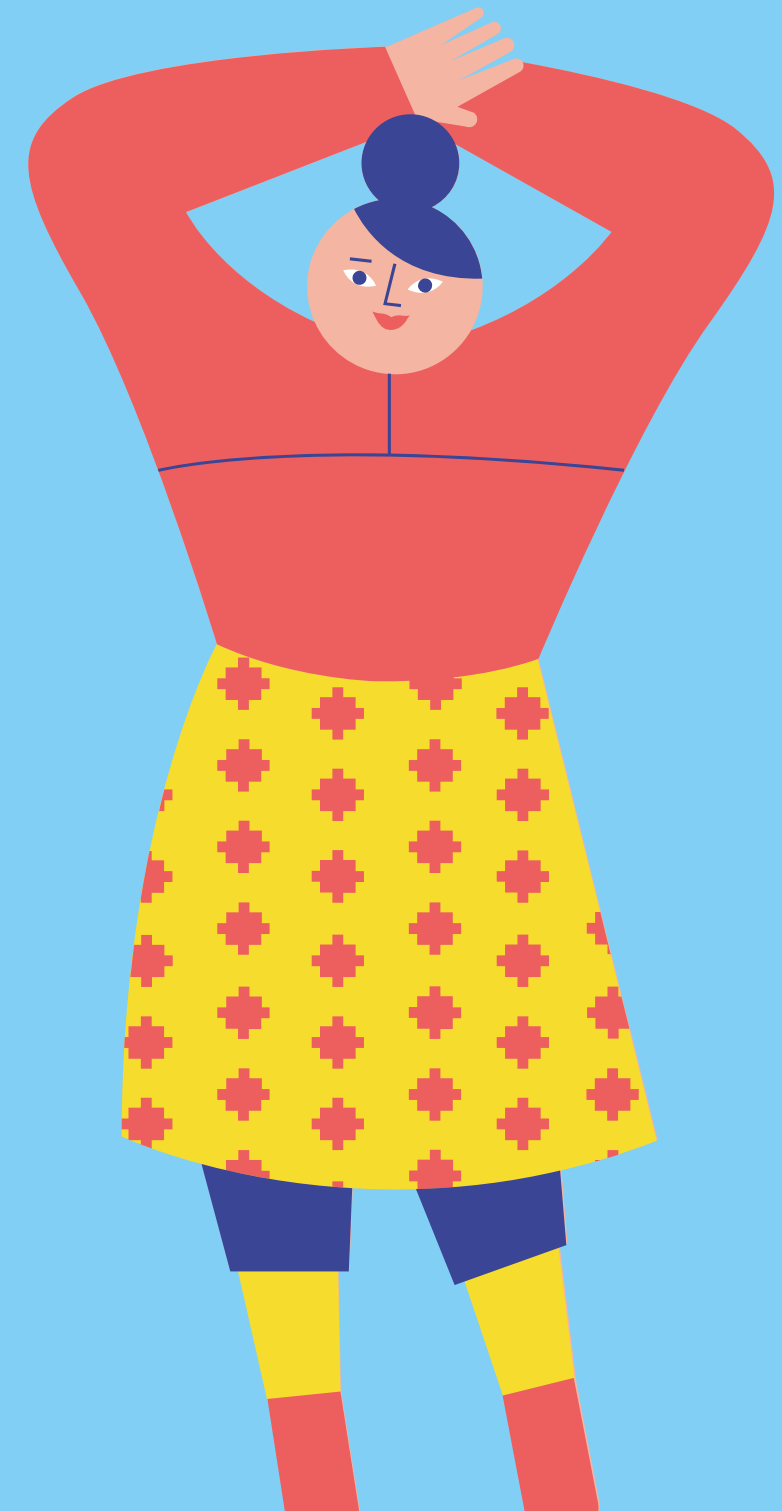
# Specific heat of water vapor

$$1.88 \frac{\text{kJ}}{\text{kg}^\circ\text{C}} \approx 2$$



# Solar constant

$$1367 \text{ W/m}^2$$



# Atmospheric

# pressure at sea

# level

/ atm

101325 Pa



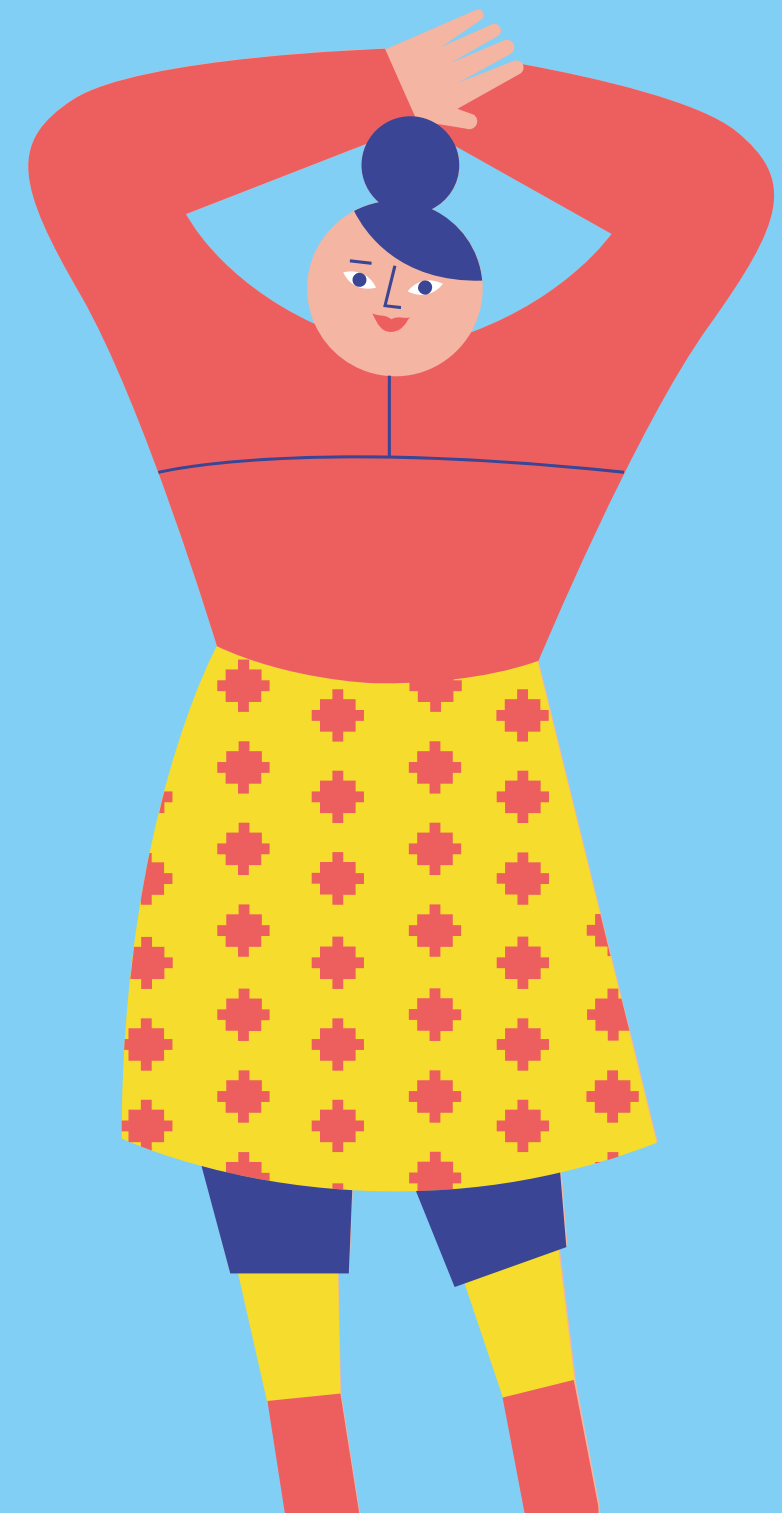
# Standard temperature

273.15 K



# R of steam

461 ~~J~~ / kg · K



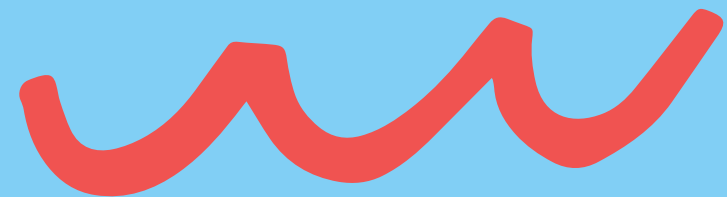
# Specific heat of steam



2.01 J/gK

# Blackbody emissivity

$$\epsilon_{\lambda}$$



Latent heat of  
sublimation (ice to  
vapor)  $2836 \frac{\text{KJ}}{\text{kg}}$



Thank you for participating in  
our icebreaker session,

Engineers!!!

